

CHAPTER 3

Inference

Until now, we have not found it necessary to assume any distributional form for the errors ε . However, if we want to make any confidence intervals or perform any hypothesis tests, we will need to do this. The common assumption is that the errors are normally distributed. In practice, this is often, although not always, a reasonable assumption. We have already assumed that the errors are independent and identically distributed (i.i.d.) with mean 0 and variance σ^2 , so we have $\varepsilon \sim N(0, \sigma^2 I)$. Now since $y = X\beta + \varepsilon$, we have $y \sim N(X\beta, \sigma^2 I)$, which is a compact description of the regression model. From this we find, using the fact that linear combinations of normally distributed values are also normal, that:

$$\hat{\beta} = (X^T X)^{-1} X^T y \sim N(\beta, (X^T X)^{-1} \sigma^2)$$

3.1 Hypothesis Tests to Compare Models

Given several predictors for a response, we might wonder whether all are needed. Consider a larger model, Ω , and a smaller model, ω , which consists of a subset of the predictors that are in Ω . If there is not much difference in the fit, we would prefer the smaller model on the principle that simpler explanations are preferred. On the other hand, if the fit of the larger model is appreciably better, we will prefer it. We will take ω to represent the null hypothesis and Ω to represent the alternative. A geometrical view of the problem may be seen in Figure 3.1.

$RSS_{\omega} - RSS_{\Omega}$ is small, then the fit of the smaller model is almost as good as the larger model and so we would prefer the smaller model on the grounds of simplicity. On the other hand, if the difference is large, then the superior fit of the larger model would be preferred. This suggests that something like:

$$\frac{RSS_{\omega} - RSS_{\Omega}}{RSS_{\Omega}}$$

would be a potentially good test statistic where the denominator is used for scaling purposes.

As it happens, the same test statistic arises from the likelihood-ratio testing approach. We give an outline of the development: If $L(\beta, \sigma^2 | y)$ is the likelihood function, then the likelihood-ratio statistic is:

$$\frac{\max_{\beta, \sigma^2 \in \Omega} L(\beta, \sigma^2 | y)}{\max_{\beta, \sigma^2 \in \omega} L(\beta, \sigma^2 | y)}$$

The test should reject if this ratio is too large. Working through the details, we find that for each model:

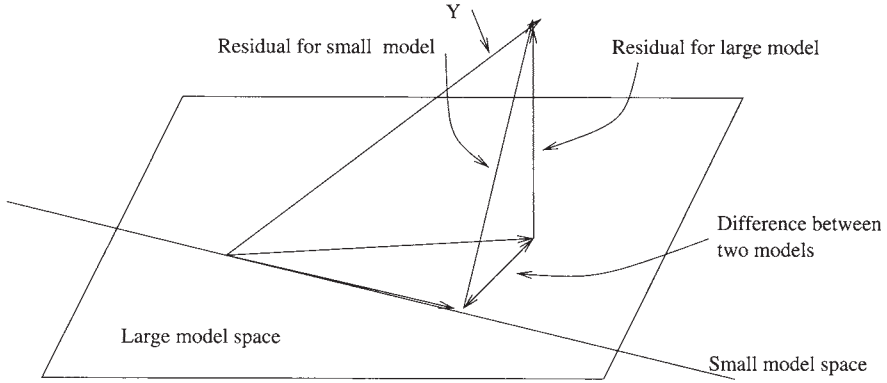


Figure 3.1 Geometrical view of the comparison between big model, Ω , and small model, ω . The squared length of the residual vector for the big model is RSS_{Ω} while that for the small model is RSS_{ω} . By Pythagoras' theorem, the squared length of the vector connecting the two fits is $RSS_{\omega} - RSS_{\Omega}$. A small value for this indicates that the small model fits almost as well as the large model and thus might be preferred due to its simplicity.

$$L(\hat{\beta}, \hat{\sigma}^2 y) \propto \hat{\sigma}^{-n}$$

which after some manipulation gives us a test that rejects if:

$$\frac{RSS_{\omega} - RSS_{\Omega}}{RSS_{\Omega}} > \text{a constant}$$

which is the same statistic suggested by the geometrical view. It remains for us to discover the null distribution of this statistic.

Now suppose that the dimension (or number of parameters) of Ω is p and the dimension of ω is q , then:

$$F = \frac{(RSS_{\omega} - RSS_{\Omega}) / (p - q)}{RSS_{\Omega} / (n - p)} \sim F_{p-q, n-p}$$

Details of the derivation of this statistic may be found in more theoretically oriented texts such as Sen and Srivastava (1990).

Thus we would reject the null hypothesis if $F > F_{p-q, n-p}^{(\alpha)}$. The degrees of freedom of a model are (usually) the number of observations minus the number of parameters so this test statistic can also be written:

$$F = \frac{(\text{RSS}_\omega - \text{RSS}_\Omega) / (df_\omega - df_\Omega)}{\text{RSS}_\Omega / df_\Omega}$$

where $df_\Omega = n - p$ and $df_\omega = n - q$. The same test statistic applies not just to when ω is a subset of Ω , but also to a subspace. This test is very widely used in regression and analysis of variance. When it is applied in different situations, the form of test statistic may be reexpressed in various different ways. The beauty of this approach is you only need to know the general form. In any particular case, you just need to figure out which models represent the null and alternative hypotheses, fit them and compute the test statistic. It is very versatile.

3.2 Testing Examples

Test of all the predictors

Are any of the predictors useful in predicting the response? Let the full model (Ω) be $y = X\beta + \epsilon$ where X is a full-rank $n \times p$ matrix and the reduced model (ω) be $y = \mu\epsilon$. We would estimate μ by $\bar{y}$. We write the null hypothesis as:

$$H_0: \beta_1 = \dots \beta_{p-1} = 0$$

Now $\text{RSS}_\Omega = (y - X\hat{\beta})^T (y - X\hat{\beta}) = \hat{\epsilon}^T \hat{\epsilon}$, the residual sum of squares for the full model, while $\text{RSS}_\omega = (y - \bar{y})^T (y - \bar{y}) = TSS$, which is sometimes known as the sum of squares corrected for the mean. So in this case:

$$F = \frac{(TSS - \text{RSS}) / (p - 1)}{\text{RSS} / (n - p)}$$

We would now refer to $F_{p-1, n-p}$ for a critical value or a p-value. Large values of F would indicate rejection of the null. Traditionally, the information in the above test is presented in an *analysis of variance table*. Most computer packages produce a variant on this. See Table 3.1 for the layout. It is not really necessary to specifically compute all the elements of the table. As the originator of the table, Fisher said in 1931, it is “nothing but a convenient way of arranging the arithmetic.” Since he had to do his calculations by hand, the table served a necessary purpose, but it is not essential now.

Source	Deg. of Freedom	Sum of Squares	Mean Square	F
Regression	$P-1$	SS_{reg}	$SS_{reg} / (p-1)$	F

Residual	$n-p$	RSS	$RSS/(n-p)$
Total	$n-1$	TSS	

Table 3.1 *Analysis of variance table.*

A failure to reject the null hypothesis is not the end of the game—you must still investigate the possibility of nonlinear transformations of the variables and of outliers which may obscure the relationship. Even then, you may just have insufficient data to demonstrate a real effect, which is why we must be careful to say “fail to reject” the null rather than “accept” the null. It would be a mistake to conclude that no real relationship exists. This issue arises when a pharmaceutical company wishes to show that a proposed generic replacement for a brand-named drug is equivalent. It would not be enough in this instance just to fail to reject the null. A higher standard would be required.

When the null is rejected, this does not imply that the alternative model is the best model. We do not know whether all the predictors are required to predict the response or just some of them. Other predictors might also be added or existing predictors transformed or recombined. Either way, the overall F -test is just the beginning of an analysis and not the end.

Let’s illustrate this test and others using an old economic dataset on 50 different countries. These data are averages from 1960 to 1970 (to remove business cycle or other short-term fluctuations). dpi is per capita disposable income in U.S. dollars; ddpi is the percentage rate of change in per capita disposable income; sr is aggregate personal saving divided by disposable income. The percentage of population under 15 (pop15) and over 75 (pop75) is also recorded. The data come from Belsley, Kuh, and Welsch (1980). Take a look at the data:

```
> data(savings)
> savings
```

	sr	pop15	pop75	dpi	ddpi
Australia	11.43	29.35	2.87	2329.68	2.87
Austria	12.07	23.32	4.41	1507.99	3.93
...cases deleted...					
Malaysia	4.71	47.20	0.66	242.69	5.08

First, consider a model with all the predictors:

```
> g <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
> summary(g)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  28.566087   7.354516    3.88  0.00033
pop15        -0.461193   0.144642   -3.19  0.00260
pop75        -1.691498   1.083599   -1.56  0.12553
dpi          -0.000337   0.000931   -0.36  0.71917
ddpi          0.409695   0.196197    2.09  0.04247
Residual standard error: 3.8 on 45 degrees of freedom
Multiple R-Squared: 0.338, Adjusted R-squared: 0.28
F-statistic: 5.76 on 4 and 45 DF, p-value: 0.00079
```

We can see directly the result of the test of whether any of the predictors have significance in the model. In other words, whether $\beta_1=\beta_2=\beta_3=\beta_4=0$. Since the p-value, 0.00079, is so small, this null hypothesis is rejected.

We can also do it directly using the F -testing formula:

```
> g <- lm (sr ~ pop15 + pop75 + dpi + ddpi, savings)
> (tss <- sum((savings$sr-mean (savings$sr))^2))
[1] 983.63
> (rss <- deviance(g))
[1] 650.71
> df.residual(g)
[1] 45
> (fstat <- ((tss-rss)/4)/(rss/df.residual(g)))
[1] 5.7557
> 1-pf (fstat, 4, df.residual (g))
[1] 0.00079038
```

Verify that the numbers match the regression summary above.

Testing just one predictor

Can one particular predictor be dropped from the model? The null hypothesis would be $H_0: \beta_i=0$. Let RSS_{Ω} be the RSS for the model with all the predictors of interest which has p parameters and let RSS_{ω} be the RSS for the model with all the same predictors except predictor i .

The F -statistic may be computed using the standard formula. An alternative approach is to use a t -statistic for testing the hypothesis:

$$t_i = \hat{\beta}_i / se(\hat{\beta}_i)$$

and check for significance using a t -distribution with $n-p$ degrees of freedom.

However, t_i^2 is exactly the F -statistic here, so the two approaches are numerically identical. The latter is less work and is presented in typical regression outputs.

For example, to test the null hypothesis that $\beta_1=0$, (that pop15 is not significant in the full model) we can simply observe that the p-value is 0.0026 from the table and conclude that the null should be rejected.

Let's do the same test using the general F -testing approach: We will need the RSS and df for the full model which are 650.71 and 45, respectively. We now fit the model that represents the null:

```
> g2 <- lm (sr ~ pop75 + dpi + ddpi, savings)
```

and compute the RSS and the F -statistic:

```
> (rss2 <- deviance (g2))
[1] 797.72
> (fstat <- (deviance (g2)-deviance (g))/
```

```
(deviance (g)/df.residual(g))
[1] 10.167
```

The p-value is then:

```
> 1-pf (fstat, 1, df.residual(g))
[1] 0.002603
```

We can relate this to the t -based test and p-value by:

```
> sqrt (fstat)
[1] 3.1885
> (tstat < - summary(g)$coef[2, 3])
[1] -3.1885
> 2 * (1-pt (sqrt (fstat), 45))
[1] 0.002603
```

A more convenient way to compare two nested models is:

```
> anova (g2, g)
Analysis of Variance Table
Model 1: sr ~ pop75 + dpi + ddpi
Model 2: sr ~ pop15 + pop75 + dpi + ddpi

Res.Df Res.Sum Sq Df Sum Sq F value Pr(>F)
1      46      798
2      45     651 1    147    10.2  0.0026
```

Understand that this test of pop15 is relative to the other predictors in the model, namely, pop75, dpi and ddpi. If these other predictors were changed, the result of the test may be different. This means that it is not possible to look at the effect of pop15 in isolation. Simply stating the null hypothesis as $H_0: \beta_{pop15} = 0$ is insufficient—information about what other predictors are included in the null is necessary. The result of the test may be different if the predictors change.

Testing a pair of predictors

Suppose we wish to test the significance of variables X_j and X_k . We might construct a table as seen above and find that both variables have p-values greater than 0.05 thus indicating that individually each one is not significant. Does this mean that both X_j and X_k can be eliminated from the model? *Not necessarily.*

Except in special circumstances, dropping one variable from a regression model causes the estimates of the other parameters to change so that we might find that after dropping X_j , a test of the significance of X_k shows that it should now be included in the model.

If you really want to check the joint significance of X_j and X_k you should fit a model with and then without them and use the general F -test discussed before. Remember that even the result of this test may depend on what other predictors are in the model.

We test the hypothesis that both pop75 and ddpi may be excluded from the model:

```

> g3 <- lm (sr ~ pop15+dpi , savings)
> anova (g3, g)
Analysis of Variance Table
Model 1: sr ~ pop15 + dpi
Model 2: sr ~ pop15 + pop75 + dpi + ddpi
Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1      47 744.12
2      45 650.71  2    9      3.41 3.2299 0.04889

```

We see that the pair of predictors is just barely significant at the 5% level.

Tests of more than two predictors may be performed in a similar way by comparing the appropriate models.

Testing a subspace

Consider this example. Suppose that y is the first year grade point average for a student, X_j is the score on the quantitative part of a standardized test and X_k is the score on the verbal part. There might also be some other predictors. We might wonder whether we need two separate scores—perhaps they can be replaced by the total, $X_j + X_k$. So if the original model was:

$$y = \beta_0 + \dots + \beta_j X_j + \beta_k X_k + \dots + \varepsilon$$

then the reduced model is:

$$y = \beta_0 + \dots + \beta_l (X_j + X_k) + \dots + \varepsilon$$

which requires that $\beta_j = \beta_k$ for this reduction to be possible. So the null hypothesis is:

$$H_0: \beta_j = \beta_k$$

This defines a linear subspace to which the general F -testing procedure applies. In our example, we might hypothesize that the effect of young and old people on the savings rate was the same or in other words that:

$$H_0: \beta_{pop15} = \beta_{pop75}$$

In this case the null model would take the form:

$$y = \beta_0 + \beta_{dep} (pop15 + pop75) + \beta_{dpi} dpi + \beta_{ddpi} ddpi + \varepsilon$$

We can then compare this to the full model as follows:

```

> g <- lm (sr ~ ., savings)
> gr <- lm (sr ~ I (pop15+pop75)+dpi+ddpi/savings)

```

```
> anova (gr, g)
Analysis of Variance Table
Model 1: sr ~ I (pop15 + pop75) + dpi + ddpi
Model 2: sr ~ pop15 + pop75 + dpi + ddpi
  Res.Df Res.Sum Sq Df Sum Sq F value Pr(>F)
1      46      674
2      45      651  1      23   1.58   0.21
```

The period in the first model formula is shorthand for all the other variables in the data frame. The function `I ( )` ensures that the argument is evaluated rather than interpreted as part of the model formula. The p-value of 0.21 indicates that the null cannot be rejected here, meaning that there is not evidence that young and old people need to be treated separately in the context of this particular model.

Suppose we want to test whether one of the coefficients can be set to a particular value. For example:

$$H_0: \beta_{ddpi} = 0.5$$

Here the null model would take the form:

$$y = \beta_0 + \beta_{pop15} pop15 + \beta_{pop75} pop75 + \beta_{dpi} dpi + 0.5 ddpi + \epsilon$$

Notice that there is now a fixed coefficient on the `ddpi` term. Such a fixed term in the regression equation is called an *offset*. We fit this model and compare it to the full:

```
> gr <- lm (sr ~ pop15+pop75+dpi+offset(0.5*ddpi), savings)
> anova (gr, g)
Analysis of Variance Table
Model 1: sr ~ pop15 + pop75 + dpi + offset(0.5 * ddpi)
Model 2: sr ~ pop15 + pop75 + dpi + ddpi
  Res.Df RSS Df Sum of Sq    F Pr(>F)
1      46 654
2      45 651  1          3 0.21  0.65
```

We see that the p-value is large and the null hypothesis here is not rejected. A simpler way to test such point hypotheses is to use a *t*-statistic:

$$t = (\hat{\beta} - c) / \text{se}(\hat{\beta})$$

where *c* is the point hypothesis. So in our example the statistic and corresponding p-value is:

```
> (tstat <- (0.409695-0.5)/0.196197)
[1] -0.46028
> 2*pt (tstat, 45)
[1] 0.64753
```

We can see the p-value is the same as before and if we square the t -statistic

```
> tstat^2
[1] 0.21186
```

we find we get the same F -value as above. This latter approach is preferred in practice since we do not need to fit two models but it is important to understand that it is equivalent to the result obtained using the general F -testing approach.

Can we test a hypothesis such as $H_0: \beta_j \beta_k = 1$ using our general theory? No. This hypothesis is not linear in the parameters so we cannot use our general method. We would need to fit a nonlinear model and that lies beyond the scope of this book.

3.3 Permutation Tests

We can put a different interpretation on the hypothesis tests we are making. For the Galapagos dataset, we might suppose that if the number of species had no relation to the five geographic variables, then the observed response values would be randomly distributed between the islands without relation to the predictors. The F -statistic is a good measure of the association between the predictors and the response with larger values indicating stronger associations. We might then ask what the chance would be under this assumption that an F -statistic would be observed as large, or larger than the one we actually observed. We could compute this exactly by computing the F -statistic for all possible (30!) permutations of the response variable and see what proportion exceed the observed F -statistic. This is a permutation test. If the observed proportion is small, then we must reject the contention that the response is unrelated to the predictors. Curiously, this proportion is estimated by the p-value calculated in the usual way based on the assumption of normal errors thus saving us from the massive task of actually computing the regression on all those computations. See Freedman and Lane (1983) for a discussion of these matters.

Let's see how we can apply the permutation test to the savings data. I chose a model with just pop75 and dpi so as to get a p-value for the F -statistic that is not too small (and therefore less interesting):

```
> g <- lm (sr ~ pop75+dpi,savings)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.056619   1.290435   5.47    1.7e-06
pop75         1.304965   0.777533   1.68    0.10
dpi          -0.000341   0.001013  -0.34    0.74
Residual standard error: 4.33 on 47 degrees of freedom
Multiple R-Squared:  0.102, Adjusted R-squared:  0.0642
F-statistic: 2.68 on 2 and 47 DF, p-value: 0.079
```

We can extract the F -statistic as:

```
> gs <- summary (g)
```

```
> gs$fstat
      value numdf  dendf
2.6796 2.0000 47.0000
```

The function `sample()` generates random permutations. We compute the F -statistic for 4000 randomly selected permutations and see what proportion exceeds the F -statistic for the original data

```
> fstats <- numeric(4000)
> for (i in 1:4000){
+ ge <- lm (sample(sr) ~ pop75+dpi, data=savings)
+ fstats[i] <- summary(ge)$fstat[1]
+ }
> length(fstats[fstats > 2.6796])/4000
[1] 0.07425
```

This should take less than a minute on any relatively new computer. If you repeat this, you will get a slightly different result each time because of the random selection of the permutations. So our estimated p-value using the permutation test is 0.07425, which is close to the normal theory-based value of 0.0791. We could reduce variability in the estimation of the p-value simply by computing more random permutations. Since the permutation test does not depend on the assumption of normality, we might regard it as superior to the normal theory based value. It does take longer to compute, so we might use the normal inference secure, in the knowledge that the results can also be justified with an alternative argument.

Tests involving just one predictor also fall within the permutation test framework. We permute that predictor rather than the response. Let's test the `pop75` predictor in the model. We can extract the t -statistic as:

```
>summary(g)$coef [2, 3]
[1] 1.6783
```

Now we perform 4000 permutations of `pop75` and check what fraction of the t -statistics exceeds 1.6783 in absolute value:

```
>tstats <- numeric(4000)
>for (i in 1:4000){
+ge <- lm (sr ~ sample(pop75)+dpi, savings)
+tstats[i] <- summary(ge)$coef[2,3]
+}
>mean (abs (tstats) > 1.6783)
[1] 0.10475
```

The outcome is very similar to the observed normal-based p-value of 0.10.

3.4 Confidence Intervals for β

Confidence intervals (CIs) provide an alternative way of expressing the uncertainty in our estimates. They are linked to the tests that we have already constructed. For the CIs and regions that we will consider here, the following relationship holds. For a $100(1-\alpha)\%$ confidence region, any point that lies within the region represents a null hypothesis that would not be rejected at the $100\alpha\%$ level while every point outside represents a null hypothesis that would be rejected. So, in one sense, the confidence region provides a lot more information than a single hypothesis test in that it tells us the outcome of a whole range of hypotheses about the parameter values. Of course, by selecting the particular level of confidence for the region, we can only make tests at that level and we cannot determine the p-value for any given test simply from the region. However, since it is dangerous to read too much into the relative size of p-values (as far as how much evidence they provide against the null), this loss is not particularly important.

The confidence region tells us about plausible values for the parameters in a way that the hypothesis test cannot. This makes it more valuable. As with testing, we must decide whether to form confidence regions for parameters individually or simultaneously. Simultaneous regions are preferable, but for more than two dimensions they are difficult to display and so there is still some value in computing the one-dimensional CIs.

We can consider each parameter individually, which leads to CIs taking the general form of:

$$\text{Estimate} \pm \text{Critical Value} \times \text{SE of Estimate}$$

or specifically in this case:

$$\hat{\beta}_i \pm t_{n-p}^{(\alpha/2)} \hat{\sigma} \sqrt{(X^T X)^{-1}_{ii}}$$

Alternatively, a $100(1-\alpha)\%$ confidence region for β satisfies:

$$(\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta) \leq p \hat{\sigma}^2 F_{p, n-p}^{(\alpha)}$$

These regions are ellipsoidally shaped. Because these ellipsoids live in higher dimensions, they cannot easily be visualized except for the two-dimensional case.

It is better to consider the joint CIs when possible, especially when the $\hat{\beta}$ are heavily correlated.

Consider the full model for the savings data:

```
> g <- lm(sr ~ ., savings)
```

where the output is displayed on page 28. We can construct individual 95% CIs for β_{pop75} :

```
> qt(0.975, 45)
[1] 2.0141
> c(-1.69 - 2.01 * 1.08, -1.69 + 2.01 * 1.08)
[1] -3.8608 0.4808
```

CI's have a duality with two-sided hypothesis tests as we mentioned above. Thus the interval contains zero, which indicates that the null hypothesis $H_0: \beta_{pop75}=0$ would not be rejected at the 5% level. We can see from the summary that the p-value is 12.5%—greater than 5%—confirming this point.

The CI for β_{pop75} is:

```
> c (0.41 -2.01 * 0.196, 0.41+2.01*0.196)
[1] 0.01604 0.80396
```

Because zero is in this interval, the null is rejected. Nevertheless, this CI is relatively wide in the sense that the upper limit is about 50 times larger than the lower limit. This means that we are not really that confident about what the exact effect of growth on savings really is, even though it is statistically significant. A convenient way to obtain all the univariate intervals is:

```
> confint (g)
              2.5 %      97.5 %
(Intercept) 13.7533307 43.3788424
pop15       -0.7525175 -0.1698688
pop75       -3.8739780 0.4909826
dpi         -0.0022122 0.0015384
ddpi        0.0145336 0.8048562
```

Now we construct the joint 95% confidence region for β_{pop15} and β_{pop75} . First, we load in a package for drawing confidence ellipses (which is not part of base R and so may need to be downloaded):

```
> library(ellipse)
```

and now the plot:

```
> plot(ellipse(g,c(2,3)), type="l", xlim=c(-1,0))
```

We add the origin and the point of the estimates:

```
> points(0,0)
> points(coef(g)[2], coef(g)[3], pch=18)
```

Since the origin lies outside the ellipse, we reject the hypothesis that $\beta_{pop15}=\beta_{pop75}=0$.

We mark the one-way CI on the plot for reference:

```
> abline(v=confint(g)[2,], lty=2)
> abline(h=confint(g)[3,], lty=2)
```

See the plot in Figure 3.2. Notice that these lines are not tangential to the ellipse. If the lines were moved out so that they enclosed the ellipse exactly, the CIs would be *jointly* correct.

In some circumstances, the origin could lie within both one-way CIs, but lie outside the ellipse. In this case, both one-at-a-time tests would not reject the null whereas the joint test would. The latter test would be preferred. It is also possible for the origin to lie outside the rectangle but inside the ellipse. In this case, the joint test would not reject the null whereas both one-at-a-time tests would reject. Again we prefer the joint test result.

Examine the correlation of the two predictors:

```
> cor (savings$pop15, savings$pop75)
[1] -0.90848
```

However, from the plot, we see that coefficients have a positive correlation. We can confirm this from:

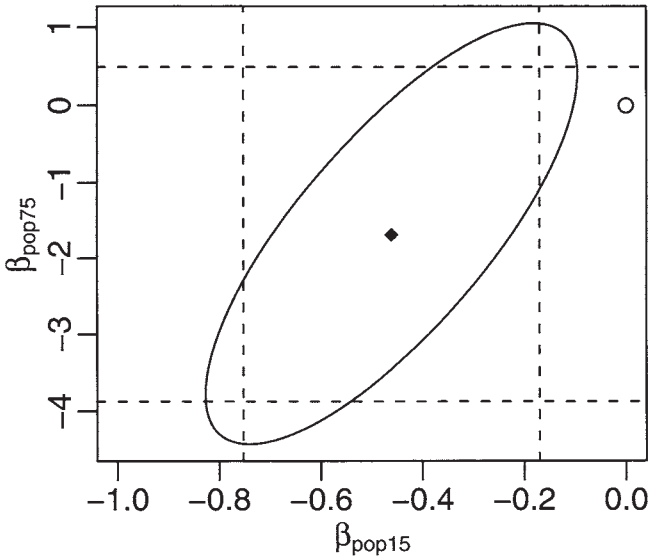


Figure 3.2 Confidence ellipse and regions for β_{pop75} and β_{pop15}

```
>summary (g, corr=TRUE) $corr
      (Intercept)      pop15      pop75      dpi      ddpi
(Intercept)  1.00000 -0.98416 -0.809111 -0.16588 -0.188265
pop15        -0.98416  1.00000  0.765356  0.17991  0.102466
pop75        -0.80911  0.76536  1.000000  -0.36705 -0.054722
```

dpi	-0.16588	0.17991	-0.367046	1.00000	0.255484
ddpi	-0.18827	0.10247	-0.054722	0.25548	1.000000

where we see that the correlation is 0.765. The correlation between predictors and the correlation between the coefficients of those predictors are often different in sign. Loosely speaking, two positively correlated predictors will attempt to perform the same job of explanation. The more work one does, the less the other needs to do and hence a negative correlation in the coefficients. For the negatively correlated predictors, as seen here, the effect is reversed.

3.5 Confidence Intervals for Predictions

Given a new set of predictors, x_0 , the predicted response is $\hat{y}_0 = x_0^T \hat{\beta}$. However, we need to assess the uncertainty in this prediction. Decision makers need more than just a point estimate to make rational choices. If the prediction has a wide CI, we need to allow for outcomes far from the point estimate. For example, suppose we need to predict the high water mark of a river. We may need to construct barriers high enough to withstand floods much higher than the predicted maximum.

We must also distinguish between predictions of the future mean response and predictions of future observations. To make the distinction clear, suppose we have built a regression model that predicts the selling price of homes in a given area that is based on predictors such as the number of bedrooms and closeness to a major highway. There are two kinds of predictions that can be made for a given x_0 :

1. Suppose a specific house comes on the market with characteristics x_0 . Its selling price will be $x_0^T \beta + \epsilon$. Since $E\epsilon=0$, the predicted price is $x_0^T \hat{\beta}$ but in assessing the variance of this prediction, we must include the variance of ϵ .
2. Suppose we ask the question—"What would a house with characteristics x_0 sell for on average?" This selling price is $x_0^T \beta$ and is again predicted by $x_0^T \hat{\beta}$ but now only the variance in $\hat{\beta}$ needs to be taken into account.

Most times, we will want the first case, which is called "prediction of a future value," while the second case, called "prediction of the mean response" is less commonly required.

We have $\text{var}(x_0^T \hat{\beta}) = x_0^T (X^T X)^{-1} x_0 \sigma^2$. A future observation is predicted to be $x_0^T \hat{\beta} + \epsilon$ (where we do not know what the future ϵ will be and we can reasonably assume this to be independent of $\hat{\beta}$). So a $100(1-\alpha) \%$ CI for a single future response is:

$$\hat{y}_0 \pm t_{n-p}^{(\alpha/2)} \hat{\sigma} \sqrt{1 + x_0^T (X^T X)^{-1} x_0}$$

If, on the other hand, you want a CI for the mean response for given x_0 , use:

$$\hat{y}_0 \pm t_{n-p}^{(\alpha/2)} \hat{\sigma} \sqrt{x_0^T (X^T X)^{-1} x_0}$$

We return to the Galápagos data for this example:

```
> g <- lm(Species ~ Area + Elevation + Nearest + Scrub + Adjacent, gala)
```

Suppose we want to predict the number of species (of tortoise) on an island with predictors 0.08, 93, 6.0, 12.0, 0.34 (same order as in the dataset). This might represent another island that was not included in the original data or we might wish to explore the effects of the predictors by experimenting with new predictor values. We can directly compute the point prediction as:

```
> x0 <- c(1, 0.08, 93, 6.0, 12.0, 0.34)
> (y0 <- sum(x0 * coef(g)))
[1] 33.92
```

This is the predicted number of species. If a whole number is preferred, we could round up to 34.

Now if we want a 95% CI for the prediction, we must decide whether we are predicting the number of species on one new island or the mean response for all islands with same predictors x_0 . Suppose that an island was not surveyed for the original dataset.

The former interval would be the one we want. For this dataset, the latter interval would be more valuable for “what if” type of calculations. First, we need the t -critical value:

```
> qt(0.975, 24)
[1] 2.0639
```

We calculate the $(X^T X)^{-1}$ matrix:

```
> x <- model.matrix(g)
> xtxi <- solve(t(x) %*% x)
```

The width of the bands for mean response CI is ($\hat{\sigma} = 60.98$):

```
> (bm <- sqrt(x0 %*% xtxi %*% x0) * 2.064 * 60.98)
[1,]
[1,] 32.89
```

and the interval is:

```
> c (y0-bm, y0+bm)
[1] 1.0296 66.8097
```

Now we compute the prediction interval for the single future response:

```
> bm <- sqrt (1+x0 %*% xtxi %*% x0) * 2.064 * 60.98
> c(y0-bm, y0+bm)
[1] -96.17 164.01
```

Of course, the number of species cannot be negative. In such instances, impossible values in the CI can be avoided by transforming the response, say taking logs (explained in a later chapter), or by using a probability model more appropriate to the response. The normal distribution is supported on the whole real line and so negative values are always possible. A better choice for this example might be the Poisson distribution which is supported on the nonnegative integers.

There is a more direct method for computing the CI. The function `predict()` requires that its second argument be a data frame with variables named in the same way as the original dataset:

```
> x0 <- data.frame (Area=0.08, Elevation=93, Nearest=6.0,
  Scruz=12, Adjacent=0.34)
> str (predict (g, x0, se=TRUE))
List of 4
 $ fit      : num 33.9
 $ se.fit   : num 15.9
 $ df       : int 24
 $ residual.scale: num 61
> predict (g, x0, interval="confidence")
      fit      lwr      upr
[1,] 33.92  1.0338 66.806
> predict (g, x0, interval="prediction")
      fit      lwr      upr
[1,] 33.92 -96.153 163.99
```

Extrapolation occurs when we try to predict the response for values of the predictor which lie outside the range of the original data. There are two different types of extrapolation:

1. Quantitative extrapolation: We must check whether the new x_0 is within the range of the original data. If not, the prediction may be unrealistic. CIs for predictions get wider as we move away from the data. We can compute these bands for the Galápagos model where we vary the Nearest variable while holding the other predictors fixed:

```
> grid <- seq (0, 100, 1)
> p <- predict (g, data.frame (Area=0.08, Elevation=93, Nearest=
  grid, Scruz=12, Adjacent=0.34), se=T, interval="confidence")
> matplot (grid, p$fit, lty=c (1,2,2), type="l",
```

```
xlab="Nearest",ylab="Species") >
rug (gala$Nearest)
```

We see that the confidence bands in Figure 3.3 become wider as we move away from the range of the data. However, this widening does not reflect the possibility that the structure of the model may change as we move into new territory. The uncertainty in the parametric estimates is allowed for, but not uncertainty about the model. The relationship may become nonlinear outside the range of the data—we have no way of knowing.

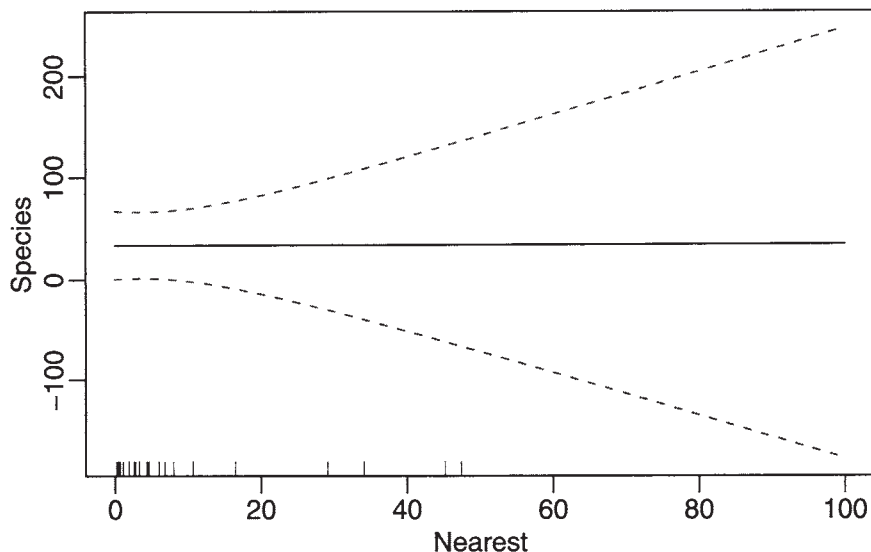


Figure 3.3 *Predicted Nearest over a range of values with 95% pointwise confidence bands for the mean response shown as dotted lines. A “rug” shows the location of the observed values of Nearest.*

2. Qualitative extrapolation: Is the new x_0 drawn from the same population from which the original sample was drawn? If the model was built in the past and is to be used for future predictions, we must make a difficult judgment as to whether conditions have remained constant enough for this to work.

3.6 Designed Experiments

In a designed experiment, the user has some control over X . For example, suppose we wish to compare several physical exercise regimes. The *experimental units* are the people we use for the study. We can choose some of the predictors such as the amount of time spent exercising or the type of equipment used. Some other predictors might not be controlled, but can be measured, such as the physical characteristics of the people. Other variables, such as the temperature in the room, might be held constant.

Our control over the conditions of the experiment allows us to make stronger conclusions from the analysis. Two important design features are *orthogonality* and *randomization*.

Orthogonality is a useful property because it allows us to more easily interpret the effect of one predictor without regard to another. Suppose we can partition X in two, $X = [X_1 \ X_2]$ such that $X_1^T X_2 = \mathbf{0}$. So now:

$$y = X\beta + \varepsilon = X_1\beta_1 + X_2\beta_2 + \varepsilon$$

and

$$X^T X = \begin{pmatrix} X_1^T X_1 & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{pmatrix} = \begin{pmatrix} X_1^T X_1 & 0 \\ 0 & X_2^T X_2 \end{pmatrix}$$

which means:

$$\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T y \quad \hat{\beta}_2 = (X_2^T X_2)^{-1} X_2^T y$$

Notice that $\hat{\beta}_1$ will be the same regardless of whether X_2 is in the model or not (and vice versa). So we can interpret the effect of X_1 without a concern for X_2 . Unfortunately, the decoupling is not perfect for suppose we wish to test $H_0: \beta_1 = 0$. We have $\text{RSS}/df = \hat{\sigma}^2$ that will be different depending on whether X_2 is included in the model or not, but the difference in F is not liable to be as large as in nonorthogonal cases.

Orthogonality is a desirable property, but will only occur when X is chosen by the experimenter. It is a feature of a good design. In observational data, we do not have direct control over X and this is the source of many of the interpretational difficulties associated with nonexperimental data.

Here is an example of an experiment to determine the effects of column temperature, gas/liquid ratio and packing height in reducing the unpleasant odor of a chemical product that was sold for household use. Read the data in and display:

```
> data (odor)
> odor
   odor  temp  gas  pack
1    66   -1  -1    0
2    39    1  -1    0
3    43   -1   1    0
4    49    1   1    0
5    58   -1   0   -1
6    17    1   0   -1
7    -5   -1   0    1
8   -40    1   0    1
9    65    0  -1   -1
10    7    0   1   -1
```

11	43	0	-1	1
12	-22	0	1	1
13	-31	0	0	0
14	-35	0	0	0
15	-26	0	0	0

The three predictors have been transformed from their original scale of measurement, for example, $\text{temp} = (\text{Fahrenheit} - 80)/40$ so the original values of the predictor were 40, 80 and 120. The data is presented in John (1971) and give an example of a *central composite* design. Here is the X -matrix:

```
> x <- as.matrix (cbind (1, odor[, -1]))
```

and $X^T X$:

```
> t (x) %*% x
      1 temp gas pack
1 15    0  0    0
temp 0    8  0    0
gas   0    0  8    0
pack  0    0  0    8
```

The matrix is diagonal. Even if temp was measured in the original Fahrenheit scale, the matrix would still be diagonal, but the entry in the matrix corresponding to temp would change. Now fit a model, while asking for the correlation of the coefficients:

```
> g <- lm (odor ~ temp + gas + pack, odor)
> summary (g, cor=T)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      15.2         9.3     1.63   0.13
temp             -12.1        12.7    -0.95   0.36
gas              -17.0        12.7    -1.34   0.21
pack             -21.4        12.7    -1.68   0.12
Residual standard error: 36 on 11 degrees of freedom
Multiple R-Squared: 0.334, Adjusted R-squared: 0.152
F-statistic: 1.84 on 3 and 11 DF, p-value: 0.199
Correlation of Coefficients:
      (Intercept) temp  gas
temp  0.00
gas   0.00         0.00
pack  0.00         0.00  0.00
```

We see that, as expected, the pairwise correlation of all the coefficients is zero. Notice that the SEs for the coefficients are equal due to the balanced design. Now drop one of the variables:

```
> g <- lm (odor ~ gas + pack, odor) >
summary (g)
Coefficients:
```

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)   15.20      9.26      1.64    0.13
gas           -17.00     12.68     -1.34    0.20
pack          -21.37     12.68     -1.69    0.12
Residual standard error: 35.9 on 12 degrees of freedom
Multiple R-Squared: 0.279, Adjusted R-squared: 0.159
F-statistic: 2.32 on 2 and 12 DF, p-value: 0.141

```

The coefficients themselves do not change, but the residual SE does change slightly, which causes small changes in the SEs of the coefficients, t -statistics and p -values, but nowhere near enough to change our qualitative conclusions.

These were data from an experiment so it was possible to control the values of the predictors to ensure orthogonality. Now consider the savings data, which are observational:

```

> g <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
> summary(g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 28.566087   7.354516   3.88  0.00033
pop15        -0.461193   0.144642  -3.19  0.00260
pop75        -1.691498   1.083599  -1.56  0.12553
dpi          -0.000337   0.000931  -0.36  0.71917
ddpi          0.409695   0.196197   2.09  0.04247
Residual standard error: 3.8 on 45 degrees of freedom
Multiple R-Squared: 0.338, Adjusted R-squared: 0.28
F-statistic: 5.76 on 4 and 45 DF, p-value: 0.00079

```

Drop pop15 from the model:

```

> g <- update(g, . ~ . - pop15)
> summary(g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 5.487494   1.427662   3.84  0.00037
pop75        0.952857   0.763746   1.25  0.21849
dpi          0.000197   0.001003   0.20  0.84499
ddpi         0.473795   0.213727   2.22  0.03162
Residual standard error: 4.16 on 46 degrees of freedom
Multiple R-Squared: 0.189, Adjusted R-squared: 0.136
F-statistic: 3.57 on 3 and 46 DF, p-value: 0.0209

```

Pay particular attention to pop75. The effect has now become positive whereas before it was negative. Granted, in both cases it is not significant, but it is not uncommon in other datasets for such sign changes to occur and for them to be significant.

Another important feature of designed experiments is the ability to randomly assign the experimental units to our chosen values of the predictors. For example, in our exercise regime experiment, we might choose an X with some desirable orthogonality properties and then fit a $y = X\beta + \varepsilon$ model. However, the subjects we use for the experiment are likely

to differ in ways that might affect the response, which could, for example, be resting heart rate. Some of these characteristics may be known and measurable such as age or weight. These can be included in X . Including such known important variables is typically beneficial, as it increases the precision of our estimates of β .

However, there will be other variables, Z that we cannot measure or may not even suspect. The model we use is $y = X\beta + \varepsilon$ while the true model might be $y = X\beta + Z\gamma + \delta$. δ is the measurement error in the response. We can assume that $E\varepsilon = 0$ without any loss of generality, because if $E\varepsilon = c$, we could simply redefine β_0 as $\beta_0 + c$ and the error would again have expectation zero. This is another reason why it is generally unwise to remove the intercept term from the model since it acts as a sink for the mean effect of unincluded variables. So we see that ε incorporates both measurement error and the effect of other variables. We will not observe Z so we cannot estimate γ . If X were orthogonal to Z , that is, $X^T Z = 0$, the estimate of β would not depend on the presence of Z . Of course, we cannot achieve this exactly, but because of the random assignment of the experimental units, which each carry an unknown value of Z , to the predictors X , we will have $\text{cor}(X, Z) = 0$. This means that the estimate of β will not be biased by Z .

Another compelling argument in favor of the randomization is that it justifies the use of permutation tests. We really can argue that any assignment of X to the experimental units was one of $n!$ possible permutations.

For observational data, no randomization can be used in assigning X to the units and orthogonality will not just happen. An unmeasured and possible unsuspected *lurking* variable Z may be the real cause of an observed relationship between y and X . For example, we will observe a positive correlation among the shoe sizes and reading abilities of elementary school students, but this relationship is driven by a lurking variable—the age of the child. In this example, the nature of Z is obvious, but in other cases, we may not be able to identify an important Z . The possible existence of such a lurking variable casts a shadow over any conclusions from observational data.

Simply using a designed experiment does not ensure success. We may choose a bad model for the data or find that the variation in the unobserved Z overwhelms the effect of the predictors. Furthermore, results from unrealistic controlled experiments in the laboratory may not extrapolate to the real world.

3.7 Observational Data

Sometimes it is not practical or ethical to collect data from a designed experiment. We cannot control the assignment of X and so we can only obtain observational data. In some situations, we can control which cases we observe from those potentially available. A *sample survey* is used to collect the data. A good survey design can allow stronger and wider conclusions, but the data will still be observational.

Interpreting models built on observational data is problematic. There are many opportunities for error and any conclusions will carry substantial unquantifiable uncertainty. Nevertheless, there are many important questions for which only observational data will ever be available and so we must make the attempt in spite of the difficulties. Suppose we fit a model to obtain the regression equation:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

What does $\hat{\beta}_1$ mean? In some cases, a β might represent a real physical constant. For example, we might attach weights x to a spring and measure the extension y . $\hat{\beta}_1$ will estimate a physical property of the spring. Such examples are rare and so usually the statistical model is just a convenience for representing a complex reality and the real meaning of a particular β is not obvious.

Let's start with the simplest interpretation: "A unit change in x_1 will produce a change of $\hat{\beta}_1$ in the response." For example, suppose y is annual income and x_1 is years of education. We might hope that $\hat{\beta}_1$ represents the predicted change in income if a particular individual had one more year of education.

The first objection is that there may be some lurking variable Z that is the real driving force behind y that also happens to be associated with x_1 . Once Z is accounted for, there may be no relationship between x_1 and y . Unfortunately, we can usually never be certain that such a Z does not exist.

Even so, what if all relevant variables have been measured? In other words, suppose there are no unidentified lurking variables. Even then the naive interpretation does not work. Consider:

$$y = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

but suppose we change $x_2 \rightarrow x_1 + x_2$; then:

$$y = \hat{\beta}_0 + (\hat{\beta}_1 - \hat{\beta}_2)x_1 + \hat{\beta}_2(x_1 + x_2)$$

The coefficient for x_1 has changed. Interpretation cannot be done separately for each variable. This is a practical problem because it is not unusual for the predictor of interest, x_1 in this example, to be mixed up in some way with other variables like x_2 . This is the problem of *collinearity* which is explored in Section 5.3.

Let's try a new interpretation: " $\hat{\beta}_1$ is the effect of x_1 when all the other (specified) predictors are held constant."

This is better, but it too has problems. Often in practice, individual variables cannot be changed without changing others too. For example, in economics we cannot expect to change tax rates without other things changing too. Furthermore, this interpretation requires the specification of the other variables—changing which other variables are included will change the interpretation. Unfortunately, there is no simple solution.

Just to amplify this, consider the effect of pop 75 on the savings rate in the savings dataset. I will fit four different models, all including pop 75, but varying the inclusion of other variables:

```
> g <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
> summary(g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  28.566087   7.354516   3.88  0.00033
pop15        -0.461193   0.144642  -3.19  0.00260
pop75        -1.691498   1.083599  -1.56  0.12553
dpi          -0.000337   0.000931  -0.36  0.71917
ddpi          0.409695   0.196197   2.09  0.04247
Residual standard error: 3.8 on 45 degrees of freedom
Multiple R-Squared: 0.338, Adjusted R-squared: 0.28
F-statistic: 3.57 on 3 and 46 DF, p-value: 0.0209
```

It is perhaps surprising that `pop75` is not significant in this model. However, `pop75` is negatively correlated with `pop15` since countries with proportionately more younger people are likely to have relatively fewer older ones and vice versa. These two variables both measure the nature of the age distribution in a country. When two variables that represent roughly the same thing are included in a regression equation, it is not unusual for one (or even both) of them to appear insignificant even though prior knowledge about the effects of these variables might lead one to expect them to be important:

```
> g2 <- lm(sr ~ pop75 + dpi + ddpi, savings)
> summary(g2)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.487494   1.427662   3.84  0.00037
pop75        0.952857   0.763746   1.25  0.21849
dpi          0.000197   0.001003   0.20  0.84499
ddpi         0.473795   0.213727   2.22  0.03162
Residual standard error: 4.16 on 46 degrees of freedom
Multiple R-Squared: 0.189, Adjusted R-squared: 0.136
F-statistic: 3.57 on 3 and 46 DF, p-value: 0.0209
```

We note that the income variable `dpi` and `pop75` are both not significant in this model and yet one might expect both of them to have something to do with savings rates. Higher values of these variables are both associated with wealthier countries.

Let's see what happens when we drop `dpi` from the model:

```
> g3 <- lm(sr ~ pop75 + ddpi, savings)
> summary(g3)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.470      1.410     3.88  0.00033
pop75          1.073      0.456     2.35  0.02299
ddpi           0.464      0.205     2.26  0.02856
Residual standard error: 4.12 on 47 degrees of freedom
Multiple R-Squared: 0.188, Adjusted R-squared: 0.154
F-statistic: 5.45 on 2 and 47 DF, p-value: 0.00742
```

Now `pop75` is statistically significant with a positive coefficient. We try dropping

`ddpi`:

```
> g4 <- lm(sr ~ pop75, savings)
```

```
> summary(g4)
```

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)    7.152      1.248    5.73  6.4e-07
pop75          1.099      0.475    2.31  0.025
Residual standard error: 4.29 on 48 degrees of freedom
Multiple R-Squared: 0.1, Adjusted R-squared: 0.0814
F-statistic: 5.34 on 1 and 48 DF, p-value: 0.0251

```

The coefficient and p-value do not change much here due to the low correlation between `pop75` and `ddpi`. Compare the coefficients and p-values for `pop75` throughout. Notice how the sign and significance change in Table 3.2.

	No. of Preds	Sign	Significant?
	4	–	no
	3	+	no
	2	+	yes
	1	+	yes

Table 3.2 *Sign and significance of $\hat{\beta}_{pop75}$*

We see that the significance and the direction of the effect of `pop75` change according to what other variables are also included in the model. We see that no simple conclusion about the effect of `pop75` is possible. We must find interpretations for a variety of models. We certainly will not be able to make any strong causal conclusions.

In observational studies, there are steps one can take to make a stronger case for causality:

1. Try to include all relevant variables. If you have omitted an obvious important variable, critics of your study will find it easy to discount your results. A link between smoking and lung cancer was observed many years ago, but the evidence did not become overwhelming until many follow-up studies had discounted potential lurking variables. For example, smokers tend to drink more alcohol than nonsmokers. By including alcohol consumption in the model, we are able to adjust for its effect and observe any remaining effect due to smoking. Even so the possibility of a unsuspected lurking variable will always exist and we can never be absolutely sure.
2. Use nonstatistical knowledge of the physical nature of the relationship. For example, we can examine the lungs of smokers.
3. Try a variety of models and see whether a similar effect is observed. In complex data analyses involving several variables, we will usually find several models that fit the data well. Sometimes the variables included in these various models may be quite

different, but if $\hat{\beta}_1$ is similar, this will increase confidence in the strength of the conclusions.

4. Multiple studies under different conditions can help confirm a relationship.
5. In a few cases, one can infer causality from an observational study. Dahl and Moretti (2003) report from U.S. census records that parents of a single girl are 5% more likely to divorce than parents of a single boy. The gap widens for larger families of girls and boys, respectively, and is even wider in several other countries. In many observational studies we can suggest a lurking variable that drives both the predictor and the response, but here, excepting recent experimental reproductive methods, the sex of a child is a purely random matter. This observational study functions like an experimental design. So in this example, we can say that the sex of the child affects the chance of divorce. The exact mechanism or reason is unclear. Census data are so large that statistical significance is assured.

It is difficult to assess the evidence in these situations and one can never be certain. The statistician is comfortable with the uncertainty expressed by hypothesis tests and CIs, but the uncertainty associated with conclusions based on observational data cannot be quantified and is necessarily subjective.

An alternative approach to interpreting parameter estimates is to recognize that the parameters and their estimates are fictional quantities in most regression situations. The “true” values may never be known (if they even exist in the first place). Instead, concentrate on predicting future values—these may actually be observed and success can then be measured in terms of how good the predictions were.

Consider a prediction made using each of the previously mentioned four models:

```
> x0 <- data.frame(pop15=32, pop75=3, dpi=700, ddpi=3)
> predict(g, x0)
[1] 9.7267
> predict(g2, x0)
[1] 9.9055
> predict(g3, x0)
[1] 10.078
> predict(g4, x0)
[1] 10.448
```

Prediction is more stable than parameter estimation. This enables a rather cautious interpretation of $\hat{\beta}_1$. Suppose the predicted value is $\hat{y}$ for given x_1 and other given predictor values. Now suppose we *observe* x_1+1 and the same other given predictor

values; then the predicted response is increased by $\hat{\beta}_1$. Notice that we have been careful not to say that we have taken a specific individual and increased his or her x_1 by 1; rather we have observed a new individual with predictor x_1+1 . To put it another way, people with college educations earn more on average than people without, but giving a college education to someone without one will not necessarily increase his or her income by the same amount.

3.8 Practical Difficulties

We have described a linear model $y = X\beta + \varepsilon$. Provided certain assumptions are satisfied, we can estimate β test any linear hypothesis about β , construct confidence regions for β and make predictions with CIs. The theory is clear and, with a little experience, the computation becomes straightforward. However, the most difficult part of regression data analysis is ensuring that the theory is appropriate for the real application. Einstein put it well:

So far as theories of mathematics are about reality; they are not certain; so far as they are certain, they are not about reality.

Problems may arise in several areas:

Nonrandom Samples

How the data were collected directly affects what conclusions we can draw. The general theory of hypothesis testing posits a *population* from which a *sample* is drawn—the sample is our data. We want to say something about the unknown *population* values β , using estimated values $\hat{\beta}$ that are obtained from the *sample* data. Furthermore, we require that the data be generated using a *simple random sample* of the population. This sample is finite in size, while the population is infinite in size or at least so large that the sample size is a negligible proportion of the whole. For more complex sampling designs, other procedures should be applied, but of greater concern is the case when the data are not a random sample at all.

Sometimes, researchers may try to select a *representative* sample by hand. Quite apart from the obvious difficulties in doing this, the logic behind the statistical inference depends on the sample being random. This is not to say that such studies are worthless, but that it would be unreasonable to apply anything more than descriptive statistical techniques. Confidence in the conclusions from such data is necessarily suspect.

A sample of convenience is where the data are not collected according to a sampling design. In some cases, it may be reasonable to proceed as if the data were collected using a random mechanism. For example, suppose we take the first 400 people from the phone book whose names begin with the letter P. Provided there is no ethnic effect, it may be reasonable to consider this a random sample from the population defined by the entries in the phone book. Here we are assuming the selection mechanism is effectively random with respect to the objectives of the study. The data are as good as random. Other situations are less clear-cut and judgment will be required. Such judgments are easy targets for criticism. Suppose you are studying the behavior of alcoholics and advertise in the media for study subjects. It seems very likely that such a sample will be biased, perhaps in unpredictable ways. In cases such as this, a sample of convenience is clearly biased in which case conclusions must be limited to the sample itself. This situation reduces to the case where the sample is the population.

Sometimes, the sample is the complete population. In this case, one might argue that inference is not required since the population and sample values are one and the same. For both regression datasets, *gala* and *savings*, that we have considered so far, the sample is effectively the population or a large and biased proportion thereof. Permutation tests

make it possible to give some meaning to the p-value when the sample is the population or for samples of convenience although one has to be clear that one's conclusion applies only to the particular sample. Another approach that gives meaning to the p-value when the sample is the population involves the imaginative concept of “alternative worlds” where the sample/population at hand is supposed to have been randomly selected from parallel universes. This argument is definitely more tenuous.

Choice and Range of Predictors

When important predictors are not observed, the predictions may be poor or we may misinterpret the relationship between the predictors and the response.

The range and conditions under which the data are collected may limit effective predictions. It is unsafe to extrapolate too much. Carcinogen trials may apply large doses to mice. What do the results say about small doses applied to humans? Much of the evidence for harm from substances such as asbestos and radon comes from people exposed to much larger amounts than that encountered in a normal life. It is clear that workers in old asbestos manufacturing plants and uranium miners suffered from their respective exposures to these substances, but what does that say about the danger to you or me?

Model Misspecification

We make assumptions about the structural and random part of the model. For the error structure, we may assume that $\epsilon \sim N(0, \sigma^2 I)$, but this may not be true. The structural part of linear model $Ey = X\beta$ may also be incorrect. The model we use may come from different sources:

1. Physical theory may suggest a model. For example, Hooke's law says that the extension of a spring is proportional to the weight attached. Models like these usually arise in the physical sciences and engineering.
2. Experience with past data. Similar data used in the past were modeled in a particular way. It is natural to see whether the same model will work with the current data. Models like these usually arise in the social sciences.
3. No prior idea exists—the model comes from an exploration of the data.

Confidence in the conclusions from a model declines as we progress through these. Models that derive directly from physical theory are relatively uncommon so that usually the linear model can only be regarded as an approximation to a complex reality.

The inference depends on the correctness of the model we use. We can partially check the assumptions about the model, but there will always be some element of doubt. Sometimes the data may suggest more than one possible model, which may lead to contradictory results.

Most statistical theory rests on the assumption that the model is correct. In practice, the best one can hope for is that the model is a fair representation of reality. A model can be no more than a good portrait. As George Box said,

All models are wrong but some are useful.

Publication and Experimenter Bias

Many scientific journals will not publish the results of a study whose conclusions do not reject the null hypothesis. If different researchers keep studying the same relationship,

sooner or later one of them will come up with a significant effect even if one really does not exist. It is not easy to find out about all the studies with negative results so it is easy to make the wrong conclusions. The news media often jump on the results of a single study, but one should be suspicious of these singleton results. Follow-up studies are often needed to confirm an effect.

Another source of bias is that researchers have a vested interest in obtaining a positive result. There is often more than one way to analyze the data and the researchers may be tempted to pick the one that gives them the results they want. This is not overtly dishonest, but it does lead to a bias towards positive results.

Practical and Statistical Significance

Statistical significance is not equivalent to practical significance. The larger the sample, the smaller your p-values will be, so do not confuse p-values with an important predictor effect. With large datasets it will be very easy to get statistically significant results, but the actual effects may be unimportant. Would we really care that test scores were 0.1% higher in one state than another or that some medication reduced pain by 2%? CIs on the parameter estimates are a better way of assessing the size of an effect. They are useful even when the null hypothesis is not rejected, because they tell us how confident we are that the true effect or value is close to the null.

It is also important to remember that a model is usually only an approximation of underlying reality which makes the exact meaning of the parameters debatable at the very least. The precision of the statement that $\beta_1 = 0$ exactly is at odds with the acknowledged approximate nature of the model. Furthermore, it is highly unlikely that a predictor that one has taken the trouble to measure and analyze has exactly zero effect on the response. It may be small but it will not be zero.

This means that in many cases, we know the point null hypothesis is false without even looking at the data. Furthermore, we know that the more data we have, the greater the power of our tests. Even small differences from zero will be detected with a large sample. Now if we fail to reject the null hypothesis, we might simply conclude that we did not have enough data to get a significant result. According to this view, the hypothesis test just becomes a test of sample size. For this reason, we prefer CIs.

Exercises

1. For the prostate data, fit a model with `lpsa` as the response and the other variables as predictors.
 - (a) Compute 90 and 95% CIs for the parameter associated with `age`. Using just these intervals, what could we have deduced about the p-value for `age` in the regression summary?
 - (b) Compute and display a 95% joint confidence region for the parameters associated with `age` and `lbph`. Plot the origin on this display. The location of the origin on the display tells us the outcome of a certain hypothesis test. State that test and its outcome.
 - (c) Suppose a new patient with the following values arrives:

```
lccavol lweight      age    lbph      svi      lcp
1.44692 3.62301 65.00000 0.30010 0.00000 -0.79851
```

```
gleason      pgg45
7.00000 15.00000
```

Predict the lpsa for this patient along with an appropriate 95% CI.

- (d) Repeat the last question for a patient with the same values except that he or she is age 20. Explain why the CI is wider.
 - (e) In the text, we made a permutation test corresponding to the F -test for the significance of all the predictors. Execute the permutation test corresponding to the t -test for age in this model. (Hint: `{summary(g)$coef[4,3]}` gets you the t -statistic you need if the model is called `g`.)
2. For the model of the previous question, remove all the predictors that are not significant at the 5% level.
 - (a) Recompute the predictions of the previous question. Are the CIs wider or narrower? Which predictions would you prefer? Explain.
 - (b) Test this model against that of the previous question. Which model is preferred?
 3. Using the `teengamb` data, fit a model with `gamble` as the response and the other variables as predictors.
 - (a) Which variables are statistically significant?
 - (b) What interpretation should be given to the coefficient for `sex`?
 - (c) Predict the amount that a male with average (given these data) status, income and verbal score would gamble along with an appropriate 95% CI. Repeat the prediction for a male with maximal values (for this data) of status, income and verbal score. Which CI is wider and why is this result expected?
 - (d) Fit a model with just income as a predictor and use an F -test to compare it to the full model.
 4. Using the `sat` data:
 - (a) Fit a model with total sat score as the response and `expend`, `ratio` and `salary` as predictors. Test the hypothesis that $\beta_{\text{salary}} = 0$. Test the hypothesis that $\beta_{\text{salary}} = \beta_{\text{ratio}} = \beta_{\text{expend}} = 0$. Do any of these predictors have an effect on the response?
 - (b) Now add `takers` to the model. Test the hypothesis that $\beta_{\text{salary}} = 0$. Compare this model to the previous one using an F -test. Demonstrate that the F -test and t -test here are equivalent.
 5. Find a formula relating R^2 and the F -test for the regression.

